

A Bayesian truth serum for subjective data*

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Collaborators on related work-in-progress

H. Sebastian Seung (MIT), Ray Weaver (MIT)

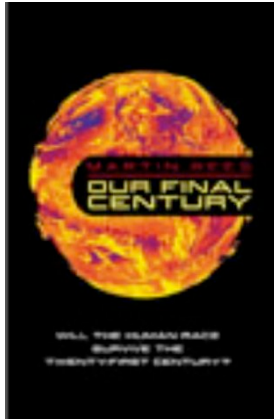
Support for related work-in-progress

NSF SES-0519141, John Simon Guggenheim Foundation,
Institute for Advanced Study

Bayesian truth serum (BTS) is a scoring instrument

- rewards truthful reporting of private opinions or judgments
- identifies experts, whose answers have ‘special status’
- designed for situations where objective truth is beyond reach
- exploits the fact that a personal opinion is a signal about the opinions of others
(the relationship between knowledge and meta-knowledge)
- analyzed under ideal conditions (rational experts, game theory)
- Distinction 1: Publicly verifiable and non-verifiable events (claims)
- Distinction 2: Rewarding individual truthfulness (“incentive compatibility”) and assessing collective truth

Sir Martin Rees, a modern Cassandra



From the BBC:

“In an eloquent and tightly argued book, Our Final Century, Sir Martin ponders the threats which face, or could face, humankind during the 21st Century. Among these, he includes natural events, such as super-eruptions and asteroid impacts, and man-made disasters like engineered viruses, nuclear terrorism and even a take-over by super-intelligent machines.”



His assessment is a sobering one:

‘I think the odds are no better than 50/50 that our present civilisation will survive to the end of the present century.’”

problem of truthfulness and truth

The truthfulness problem is to give the Cassandra a reason — a financial or reputational incentive, to voice opinions that will be greeted with disbelief.

The truth problem is to confirm that the Cassandra is genuine — that her judgment should overrule the opinions of the majority.

If judgments are verifiable then we can use prediction markets

Examples of verifiable claims:

- business forecasts
- medical forecasts
- sports forecasts
- weather forecasts
- scientific predictions

intrade: prices of Gore nominated contract



Fundamental limitation of prediction markets: They must be linked to an exact public event

Foresight Exchange Bush04 wager definition:

Fundamental limitation of prediction markets: They must be linked to an exact public event

Foresight Exchange Bush04 wager definition:

This claim will be TRUE even if elections are postponed or G.W. Bush remains in power by staging a coup.

If there are events which make it confusing who the U.S. president is, as of 2005-02-01, this claim is true if G.W. Bush is leading a sovereign government in at least part of the territory of the United States of America (as of 2001-01-01) that has recognition of at least one of the U.N. Security Council permanent members (Britain, France, China and Russia) other than the United States.

The Foresight Exchange Prediction Market

<http://www.ideosphere.com/>

Top 10 Claims by Transaction Volume in the Last 7 Days

Rank	Volume	%	Symbol	Bid/Ask/Last	Short Description
1	2581	47.5%	Gas\$3	14/ 15/ 13	US gasoline prices reach \$3.00
2	1018	18.7%	MJ06	62/ 67/ 62	Michael Jackson found guilty
3	285	5.2%	HRC08	18/ 19/ 18	Hillary Clinton US Pres by2009
4	202	3.7%	T2007	97/ 98/ 98	True on Jan 1 2007
5	160	2.9%	Marbrg	16/ 23/ 17	Marburg kills 1000 within year
6	116	2.1%	CFsn	15/ 16/ 15	Cold Fusion
7	114	2.1%	Immo	28/ 30/ 29	Immortality by 2050
8	100	1.8%	Tran	46/ 47/ 46	Machine Translation by 2015
9	100	1.8%	Trade9	48/ 50/ 50	trade deficit in 2009
10	95	1.7%	UK0505	65/ 69/ 70	Labor MP's in UK parliament

But what about actual guilt?

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Markets cannot be defined for nonverifiable claims

Examples of verifiable claims:

business forecasts
medical forecasts
sports forecasts
weather forecasts
scientific predictions

Examples of nonverifiable claims:

historical interpretations
actual guilt or innocence
remote future forecasts
artistic judgments
cultural interpretations

BTS is designed for non-verifiable content

It works at the level of one question

(i) The best current estimate of the temperature change by 2100 is (check one):

___ $\leq 2^{\circ}\text{C}$ < ___ $\leq 4^{\circ}\text{C}$ < ___ $\leq 6^{\circ}\text{C}$ < ___ $\leq 8^{\circ}\text{C}$ < ___

(ii) On current evidence, the probability that Fermat would have been able to prove Fermat's Theorem is (check one):

___ $\leq .000001$ < ___ $\leq .001$ < ___ $\leq .1$ < ___ $\leq .5$ < ___

(iii) Have you had more than twenty sexual partners over the past year?

(Yes / No)

(iv) Which wine would you take as a before-dinner drink?

(Red / White)

How it works...

How it works...

Ask each respondent r for dual reports:

(1) an endorsement of an answer to an m -multiple-choice question

$x_k^r \in \{0,1\}$ indicates whether respondent r has endorsed answer $k \in \{1,...,m\}$

(2) a prediction $(y_1^r, ..., y_m^r)$ of the sample distribution of endorsements

Then calculate BTS scores

- The score is defined relative to the reported sample averages:

$\bar{x}_k$ = fraction endorsing answer k

$\bar{y}_k$ = geometric average of endorsement predictions for answer k

- The total BTS score for person r , for endorsement $(x_1^r, \dots, x_m^r)$ and prediction $(y_1^r, \dots, y_m^r)$:

BTS score = Information score + Prediction score

$$u^r = \sum_{k=1}^m x_k^r \log \frac{\bar{x}_k}{\bar{y}_k} + \sum_{k=1}^m \bar{x}_k \log \frac{y_k^r}{\bar{x}_k}$$

The Information score measures whether an answer is surprisingly common

- The score is defined relative to the reported sample averages:

$\bar{x}_k$ = fraction endorsing answer k

$\bar{y}_k$ = geometric average of endorsement predictions for answer k

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The prediction score measures prediction accuracy
(and equals zero for a perfect prediction)

- The score is defined relative to the reported sample averages:

$\bar{x}_k$ = fraction endorsing answer k

$\bar{y}_k$ = geometric average of endorsement predictions for answer k

- The total BTS score for person r , for endorsement $(x_1^r, \dots, x_m^r)$ and prediction $(y_1^r, \dots, y_m^r)$:

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THEOREM (in English)

In a large sample, everyone expects their truthful answer to be the most surprisingly common answer

Therefore, to maximize expected score
you must tell the truth

Comparing BTS and prediction markets

- Common characteristics:
 - incentive compatible (truthtelling is optimal)
 - zero-sum (budget balance)
 - non-democratic aggregation of information, favoring informed participants (experts)
- Differences
 - BTS is one-shot, markets are dynamic
 - BTS is not restricted to verifiable events (claims)

The underlying Bayesian model (drawing from a bag containing balls of m different colors, representing m possible answers)

- Relative frequency of opinions is an unknown vector, $\omega = (\omega_1, \dots, \omega_m)$

(This is the unknown mixture of balls in the bag)

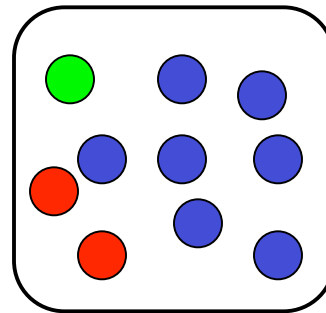
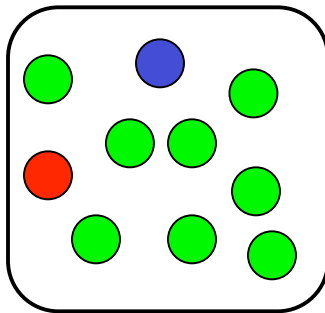
- Everyone has the same prior probability distribution $p(\omega)$ over possible relative frequencies
- Person r gets a signal $t^r \in \{1, \dots, m\}$ representing his opinion

(This is his drawing of one ball from the bag)

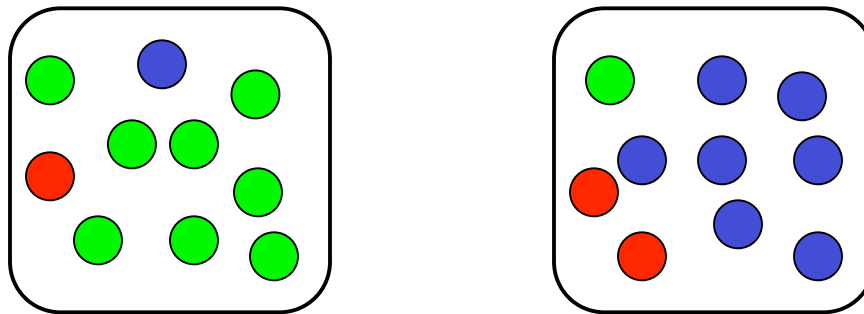
- A person r who holds opinion j treats this as a sample of one, yielding a posterior distribution $p(\omega \mid t^r=j)$ on ω , which is different for each j .
- Conditional independence:
$$p(t^r=j, t^s=k \mid \omega) = p(t^r=j \mid \omega) p(t^s=k \mid \omega)$$

A computational example

Drawing a ball (with replacement)
from one of two possible bags
The bags are a priori equally likely



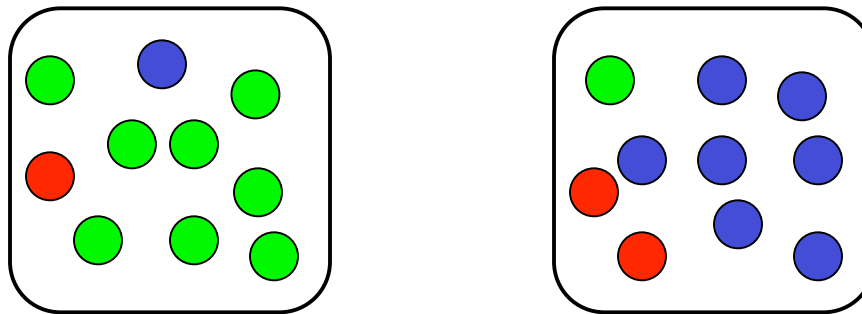
Prior expected frequencies



$$E(\bar{x}_i)$$

$i = \text{Blue}$	.40
$i = \text{Red}$	.15
$i = \text{Green}$	.45

Suppose that the ball you draw is Red

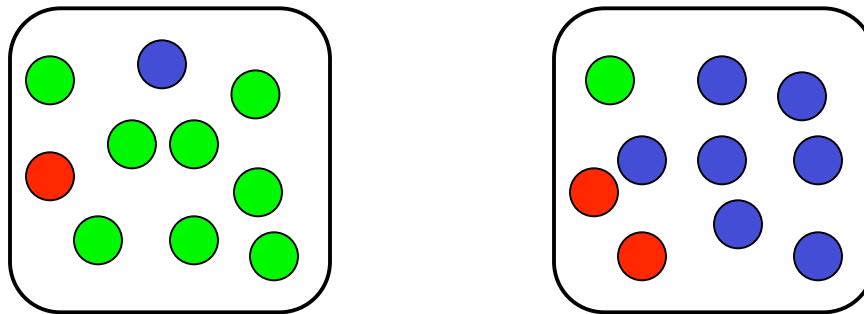


$$E(\bar{x}_i)$$

$$E(\bar{x}_i | t^r = \text{Red})$$

$i = \text{Blue}$	.40
$i = \text{Red}$	.15
$i = \text{Green}$	.45

Posterior expected frequencies, given 1 Red draw

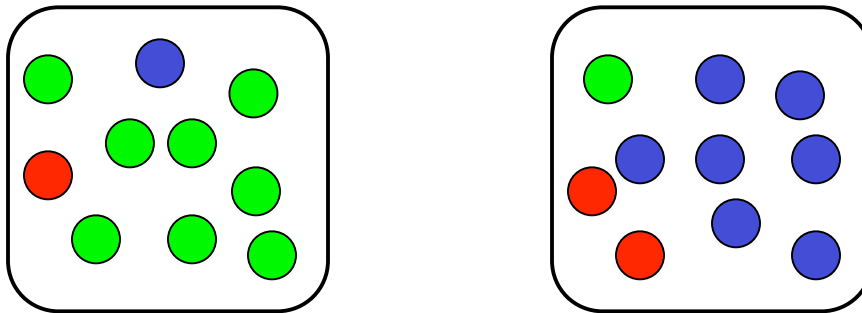


$$E(\bar{x}_i)$$

$$E(\bar{x}_i | t^r = \text{Red})$$

$i = \text{Blue}$	.40	.50
$i = \text{Red}$	.15	.17
$i = \text{Green}$	.45	.33

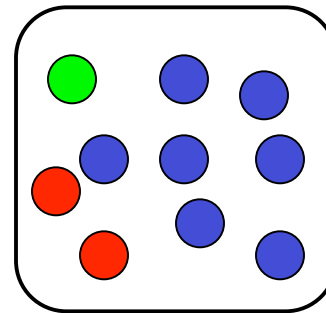
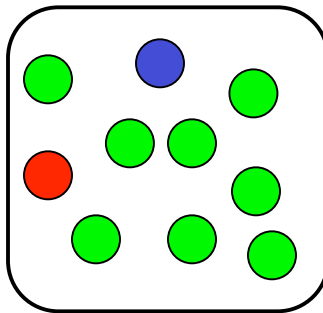
A Red draw is a more favorable
signal for Blue than for Red



$$E(\bar{x}_i) \quad E(\bar{x}_i | t^r = \text{Red})$$

$i = \text{Blue}$	.40	.50
$i = \text{Red}$	.15	.17
$i = \text{Green}$	.45	.33

Computational validation of BTS theorem



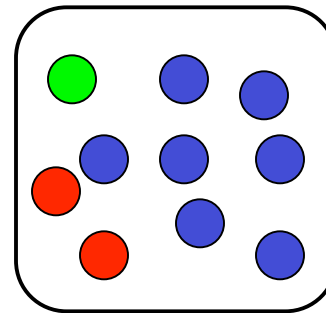
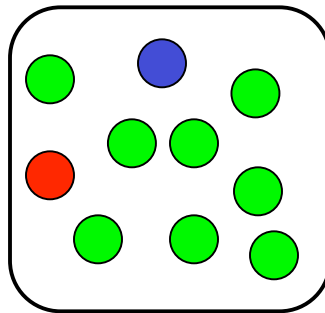
$$E(\bar{x}_i)$$

$$E(\bar{x}_i | t^r = Red)$$

$$E(\log \frac{\bar{x}_i}{\bar{y}_i} | t^r = Red)$$

$i = \text{Blue}$	.40	.50
$i = \text{Red}$	.15	.17
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Computational validation of BTS theorem



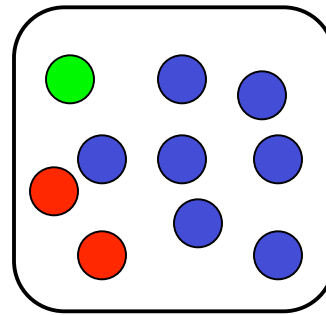
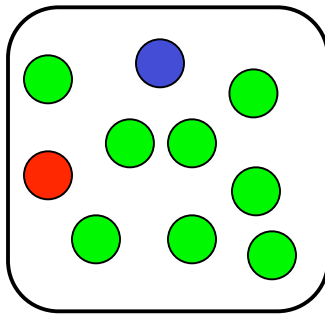
$$E(\bar{x}_i)$$

$$E(\bar{x}_i | t^r = Red)$$

$$E(\log \frac{\bar{x}_i}{\bar{y}_i} | t^r = Red)$$

$i = \text{Blue}$	.40	.50	
$i = \text{Red}$	.15	.17	
$i = \text{Green}$	.45	.33	-.48

Computational validation of BTS theorem



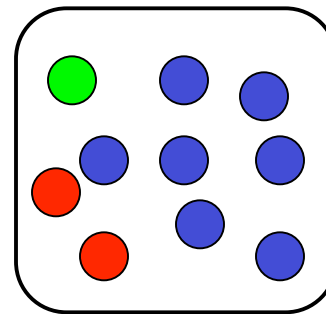
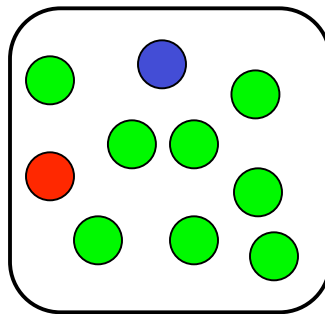
$$E(\bar{x}_i)$$

$$E(\bar{x}_i | t^r = Red)$$

$$E(\log \frac{\bar{x}_i}{\bar{y}_i} | t^r = Red)$$

$i = \text{Blue}$	.40	.50	-.06
$i = \text{Red}$	.15	.17	
$i = \text{Green}$	.45	.33	-.48

Drawing Red provides stronger evidence for Blue than for Red, but Red remains the optimal answer



	$E(\bar{x}_i)$	$E(\bar{x}_i t^r = Red)$	$E(\log \frac{\bar{x}_i}{\bar{y}_i} t^r = Red)$
$i = \text{Blue}$	.40	.50	-.06
$i = \text{Red}$	.15	.17	+.03
$i = \text{Green}$	.45	.33	-.48

Is the Bayesian model realistic?
Imagine that your host offers a glass of
white or red wine before dinner...



Which would you take?

Estimate the % that would take white ...

Your preference “wins” to the extent that it is more popular than collectively estimated



Claim:

Best strategy is to state your true preference

Typical estimates of the fraction that selects White

Estimates by those who
personally prefer White

75 %

50 %

60 %

65%

average 63 %

Estimates by those who
personally prefer Red

30%

40 %

25 %

20 %

76%

60%

average 42 %

Note the difference in average estimates...
This would be consistent with Bayesian updating*

Estimates by those who
personally prefer White

75 %
50 %
60 %
65%

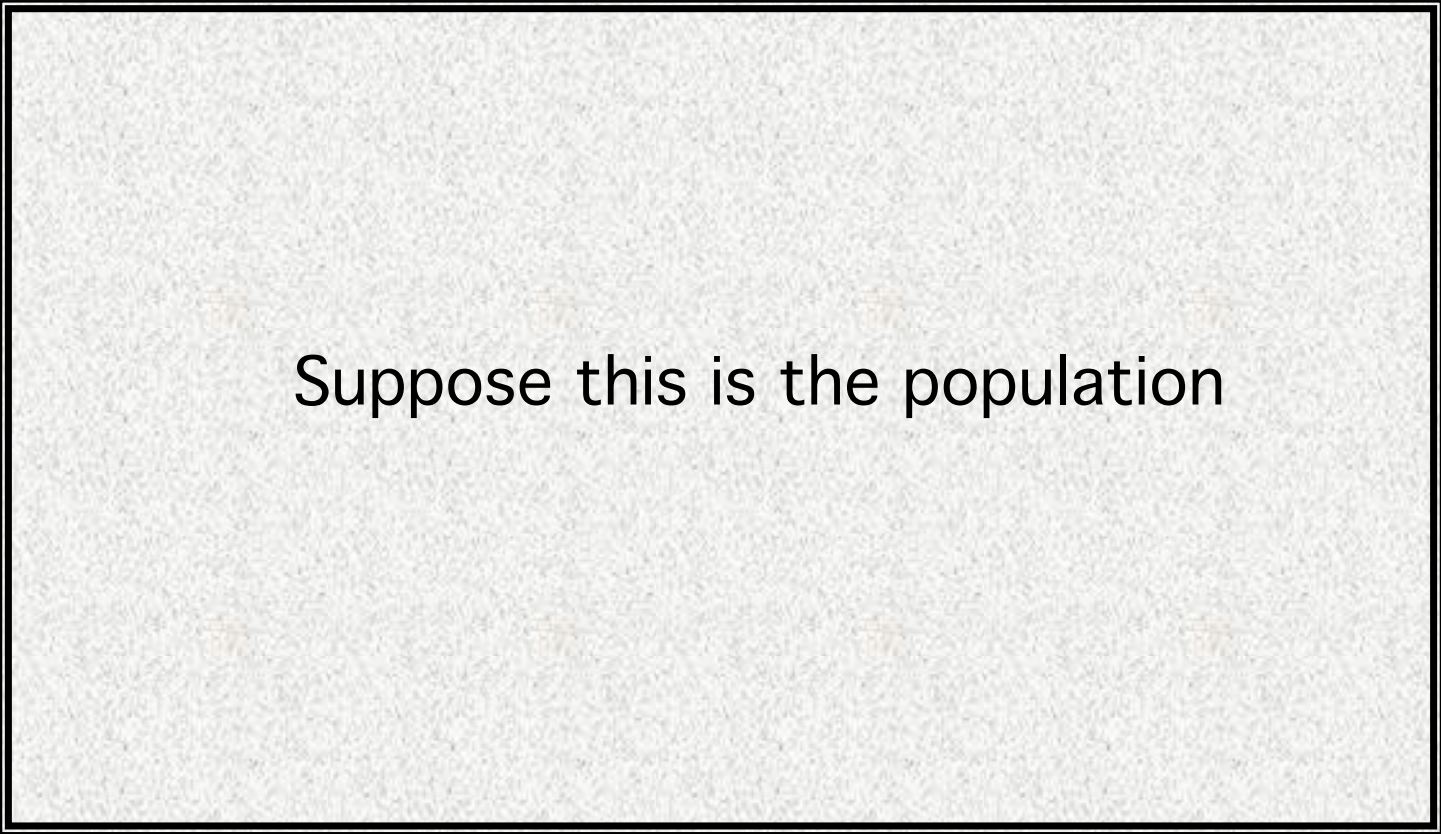
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average 63 %

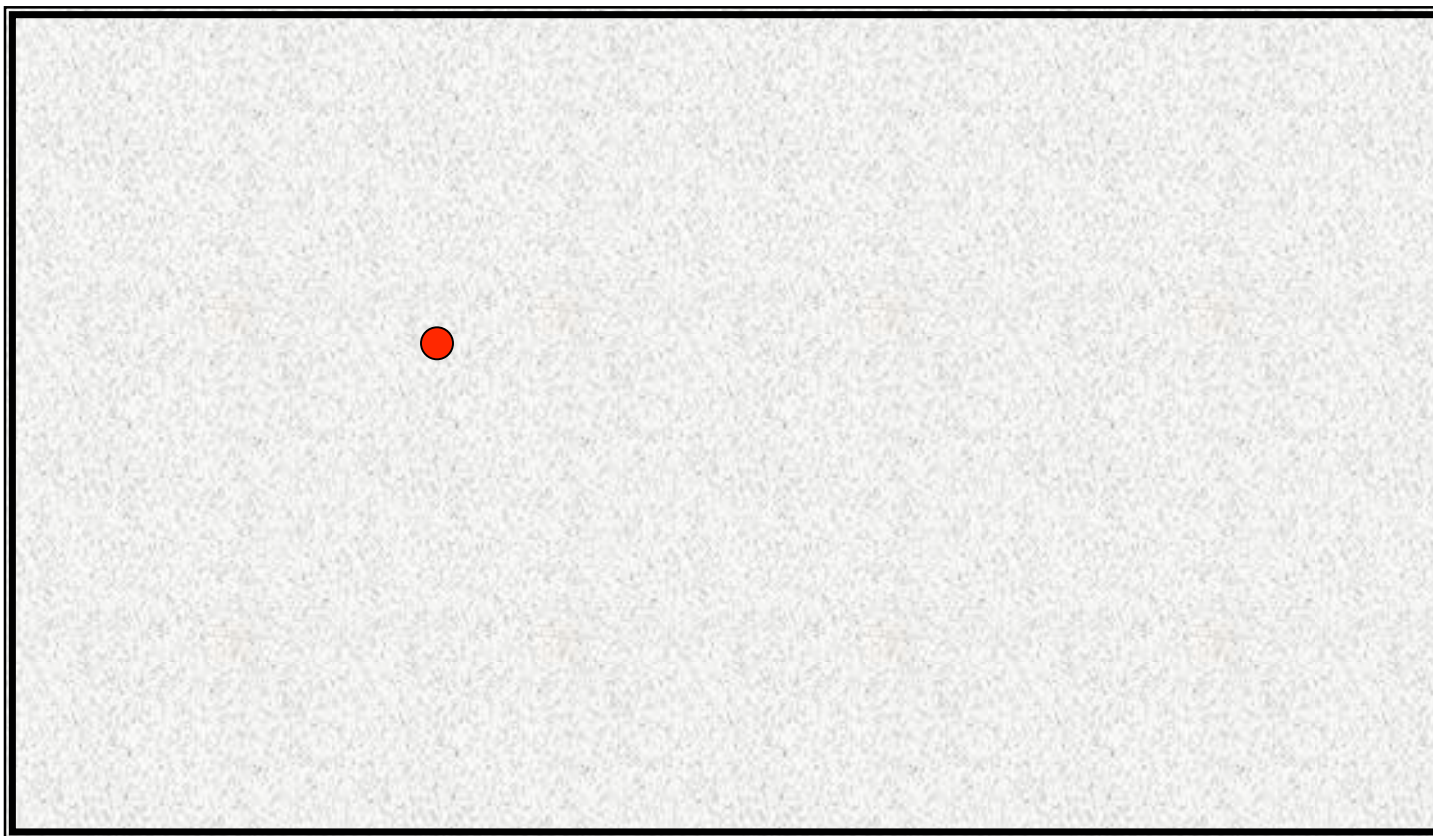
average 42 %

The intuitive argument for $m=2$

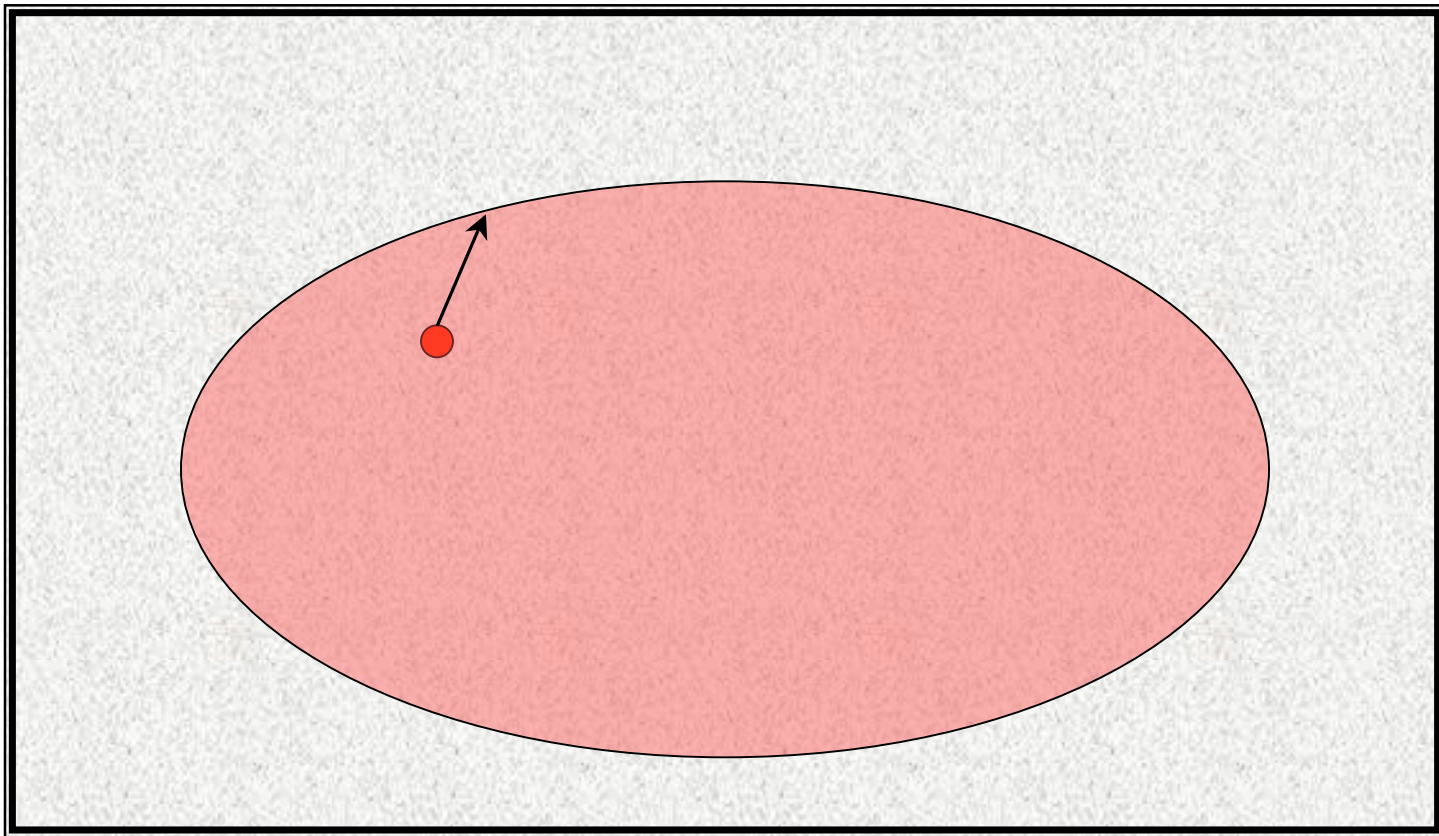


Suppose this is the population

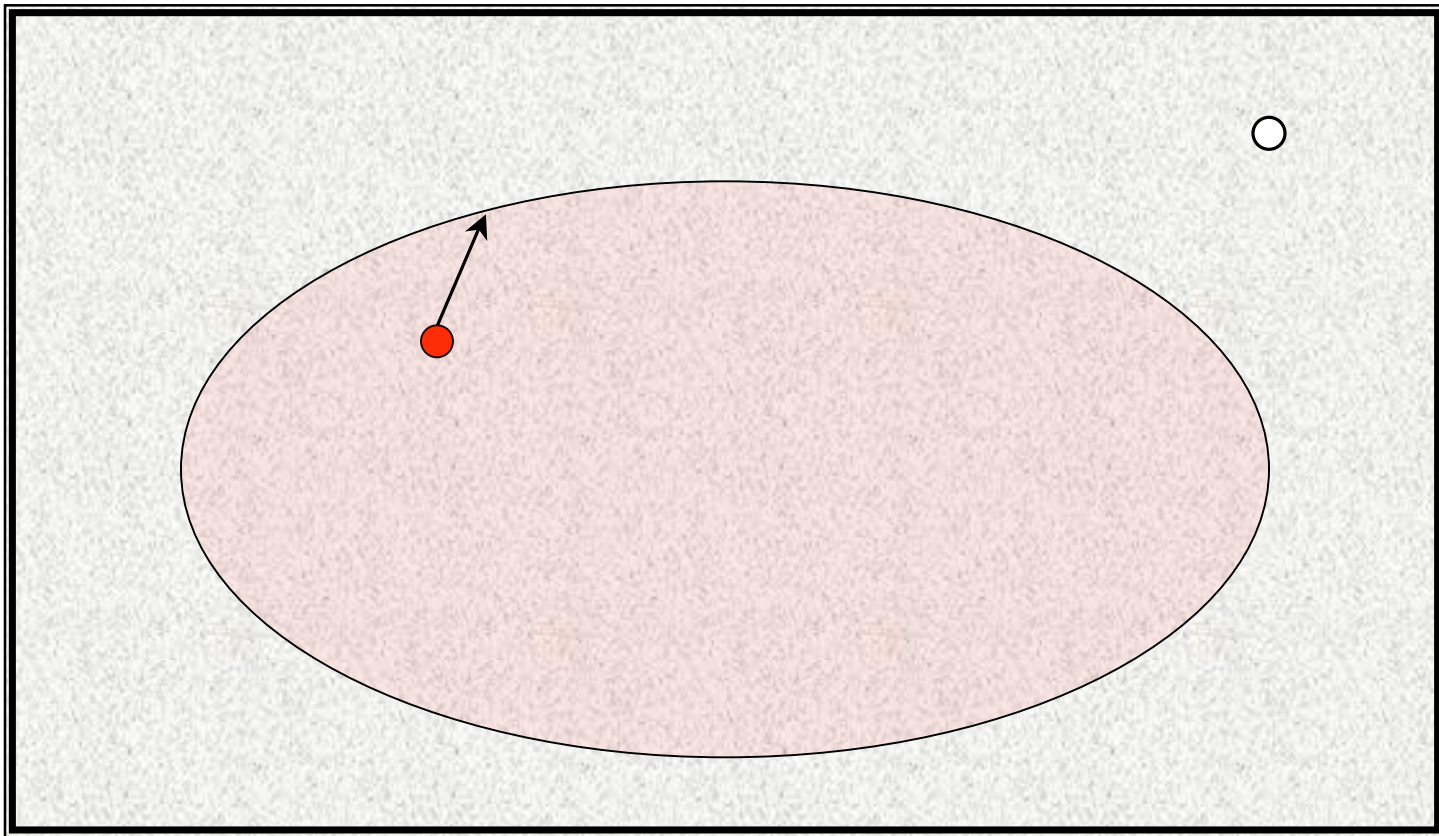
and I happen to like Red



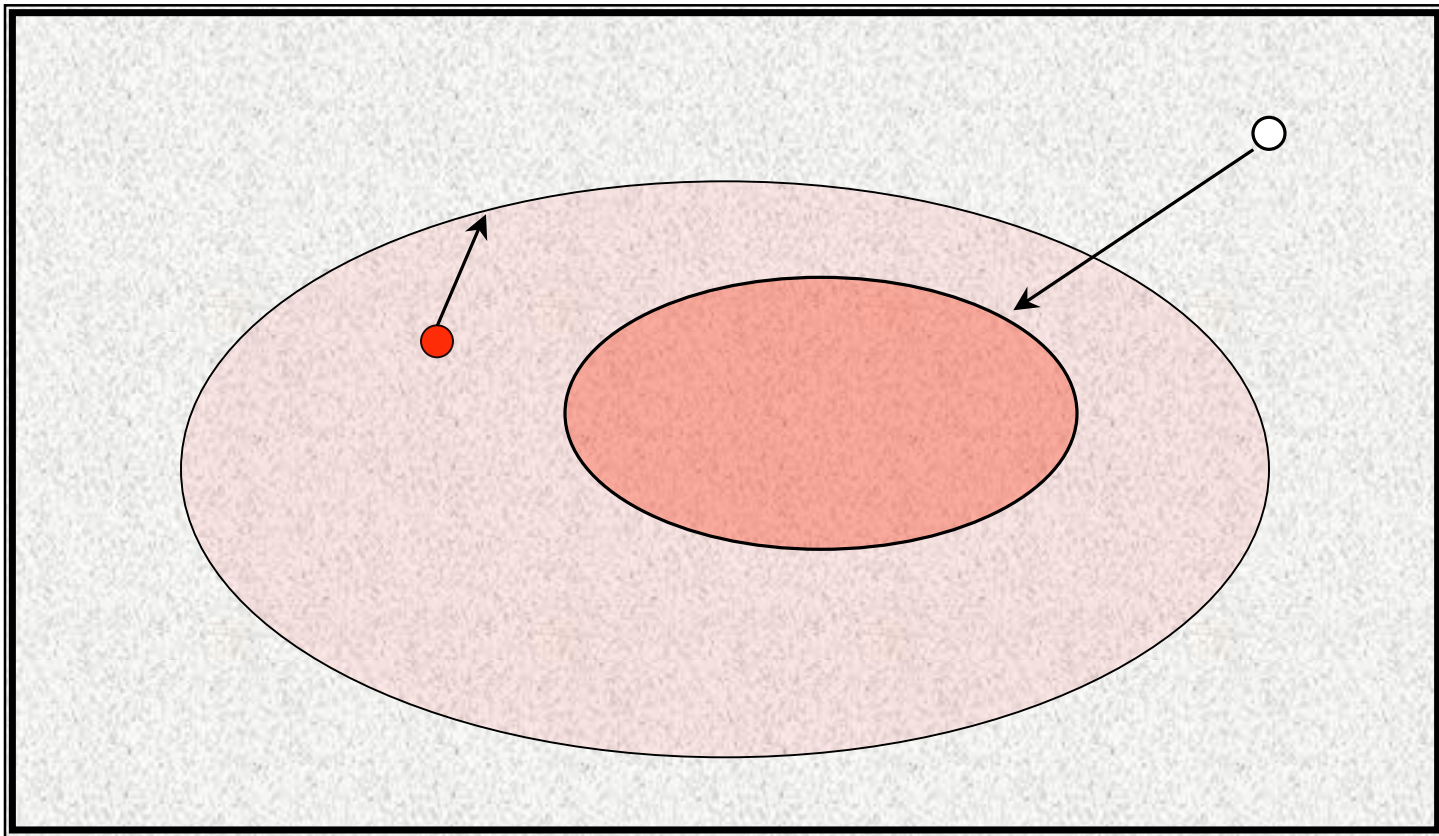
This is my best estimate of the Red share
(e.g., 50%)



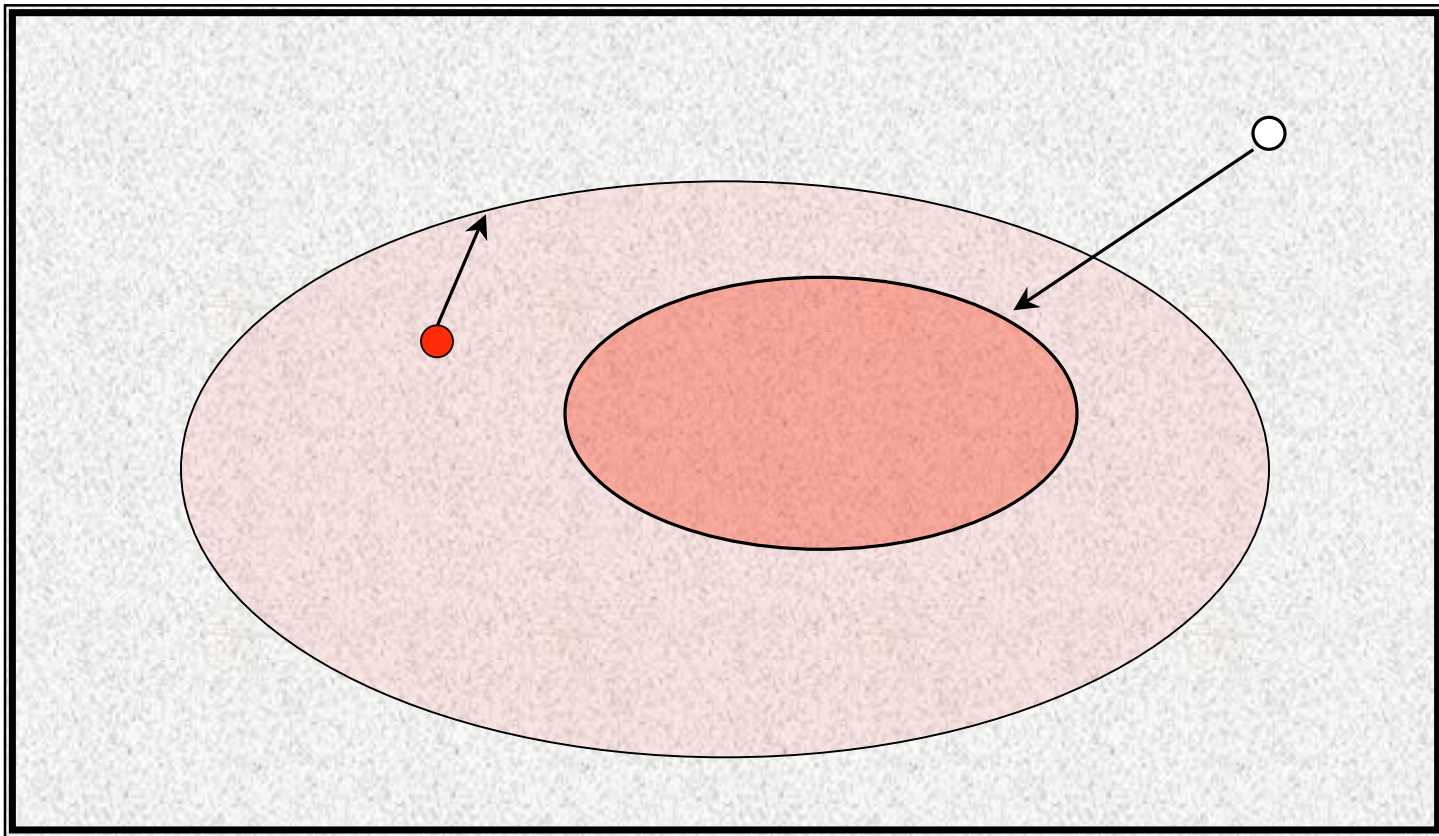
Bayesian reasoning implies that someone who likes White will estimate a smaller share for Red



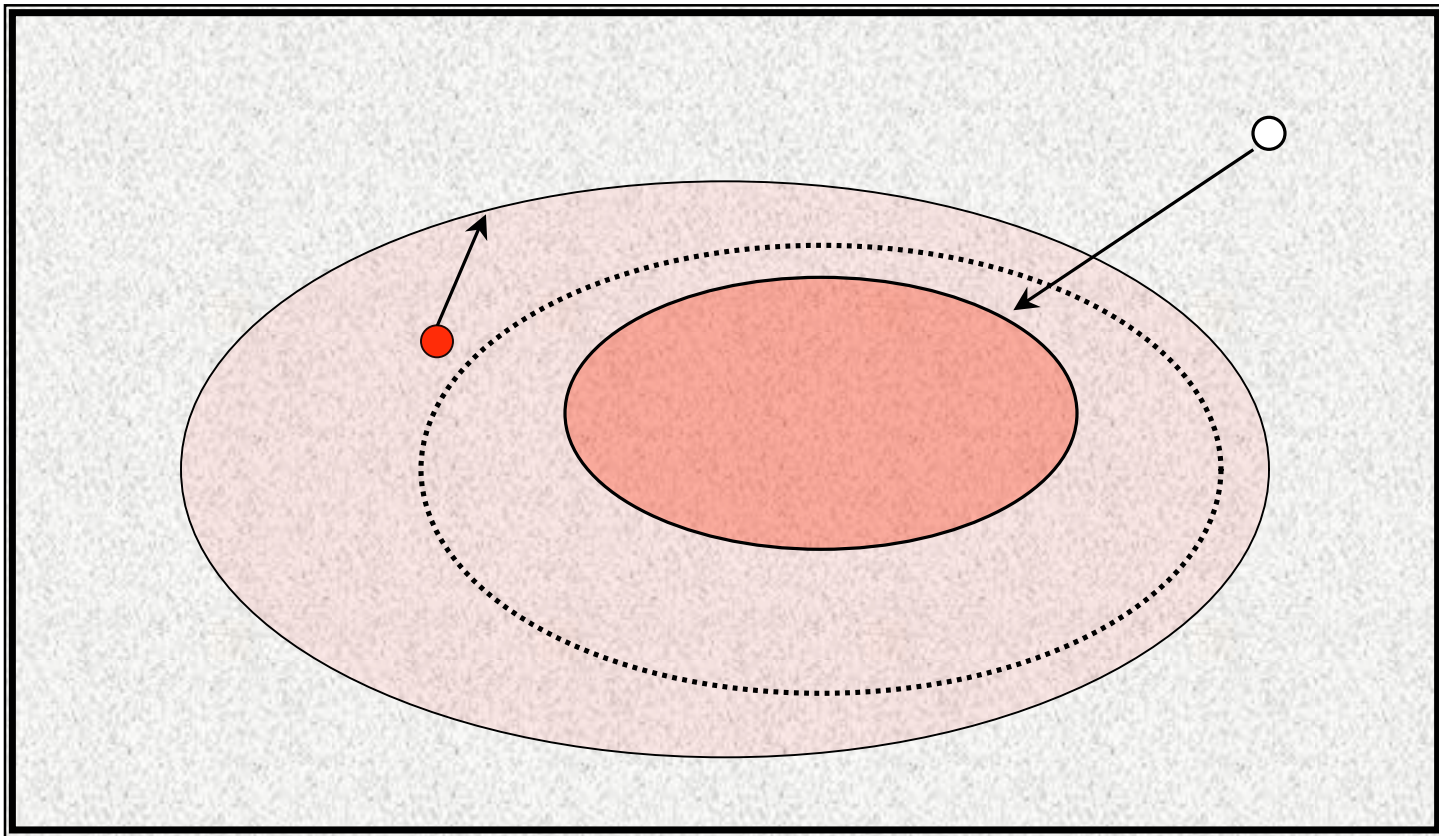
Bayesian reasoning implies that someone who likes White will estimate a smaller share for Red



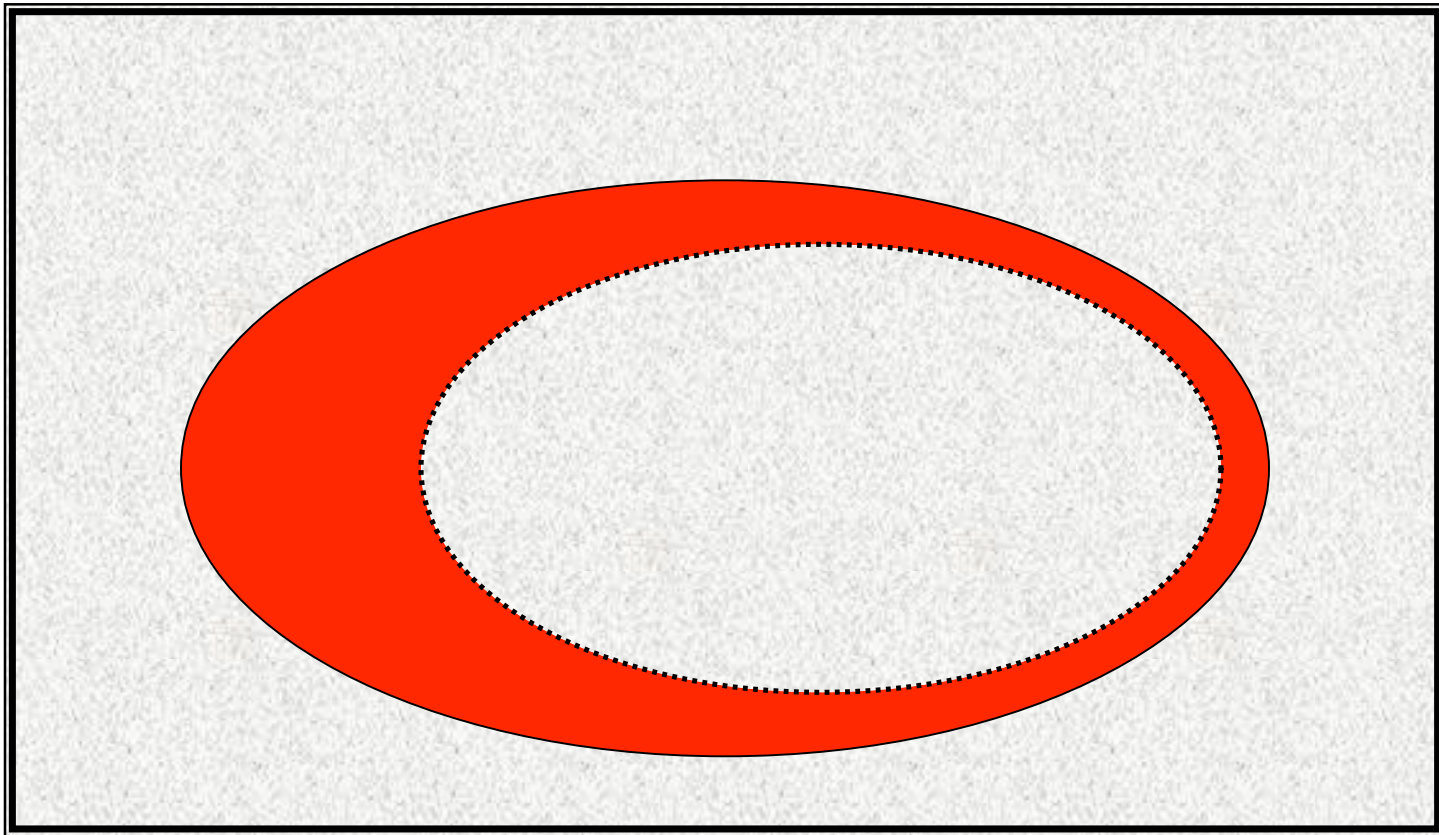
The average predicted share for Red will fall somewhere between these two estimates



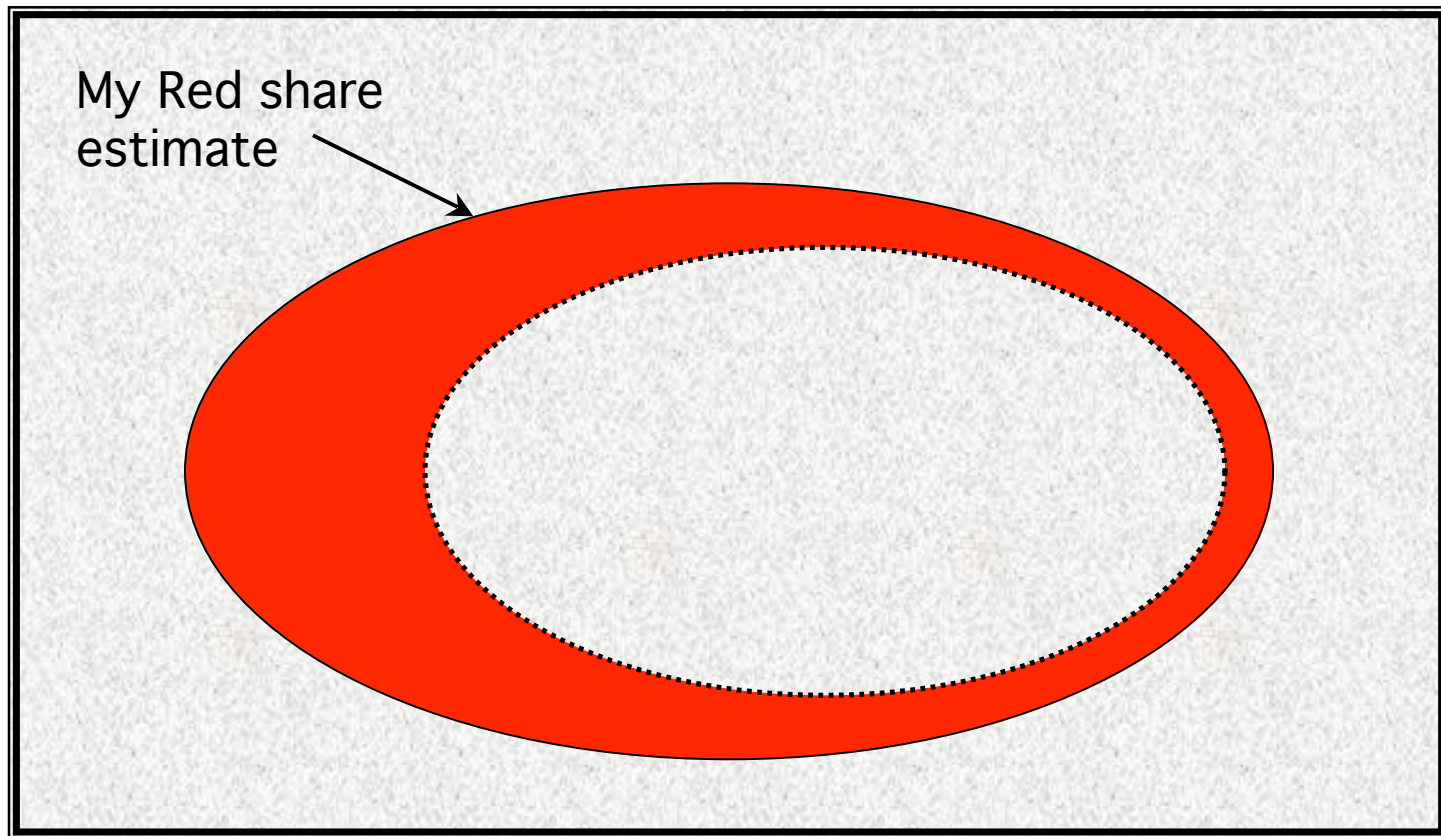
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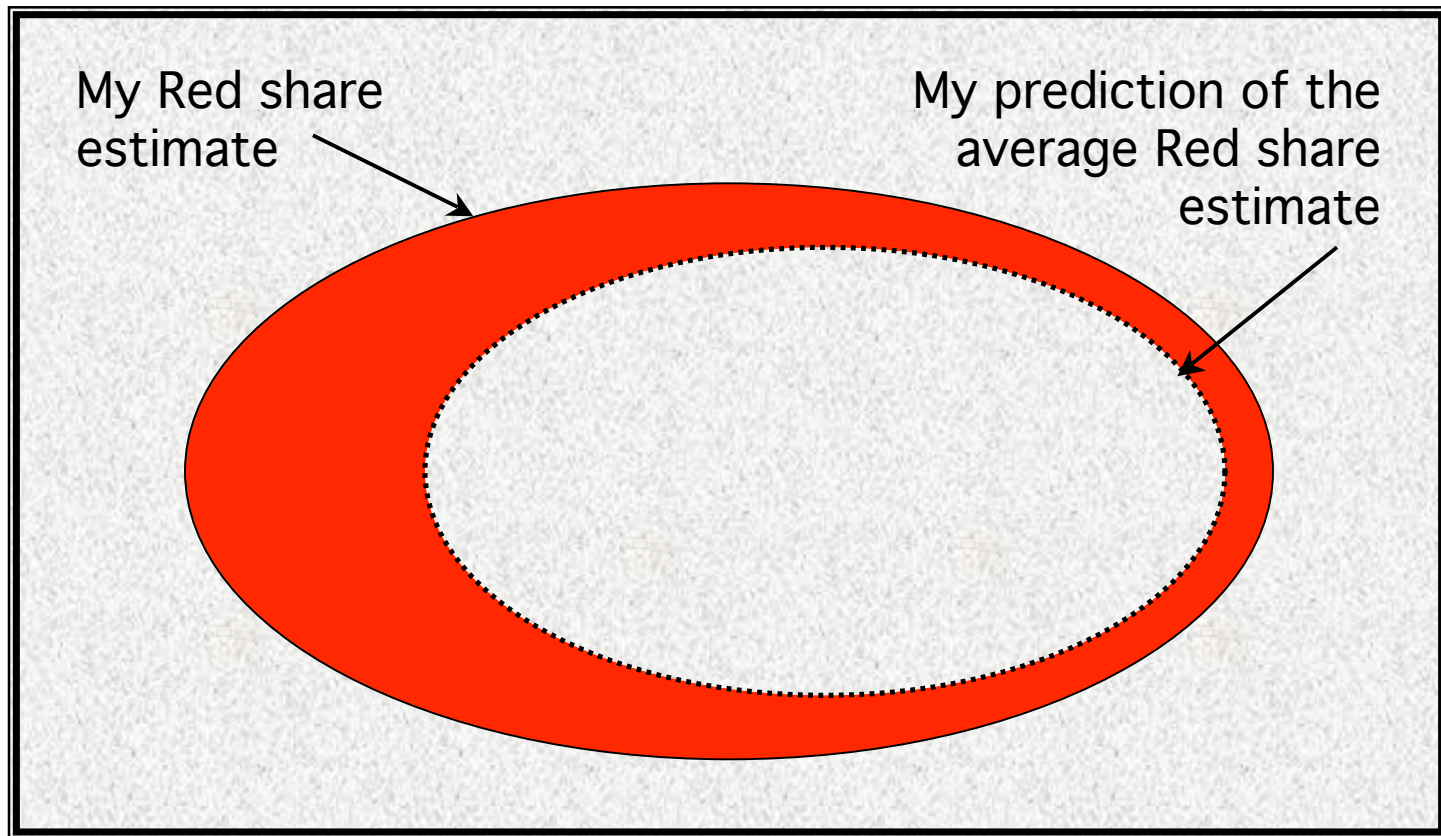
Hence, if I like Red I should believe that
the share for Red will be underestimated



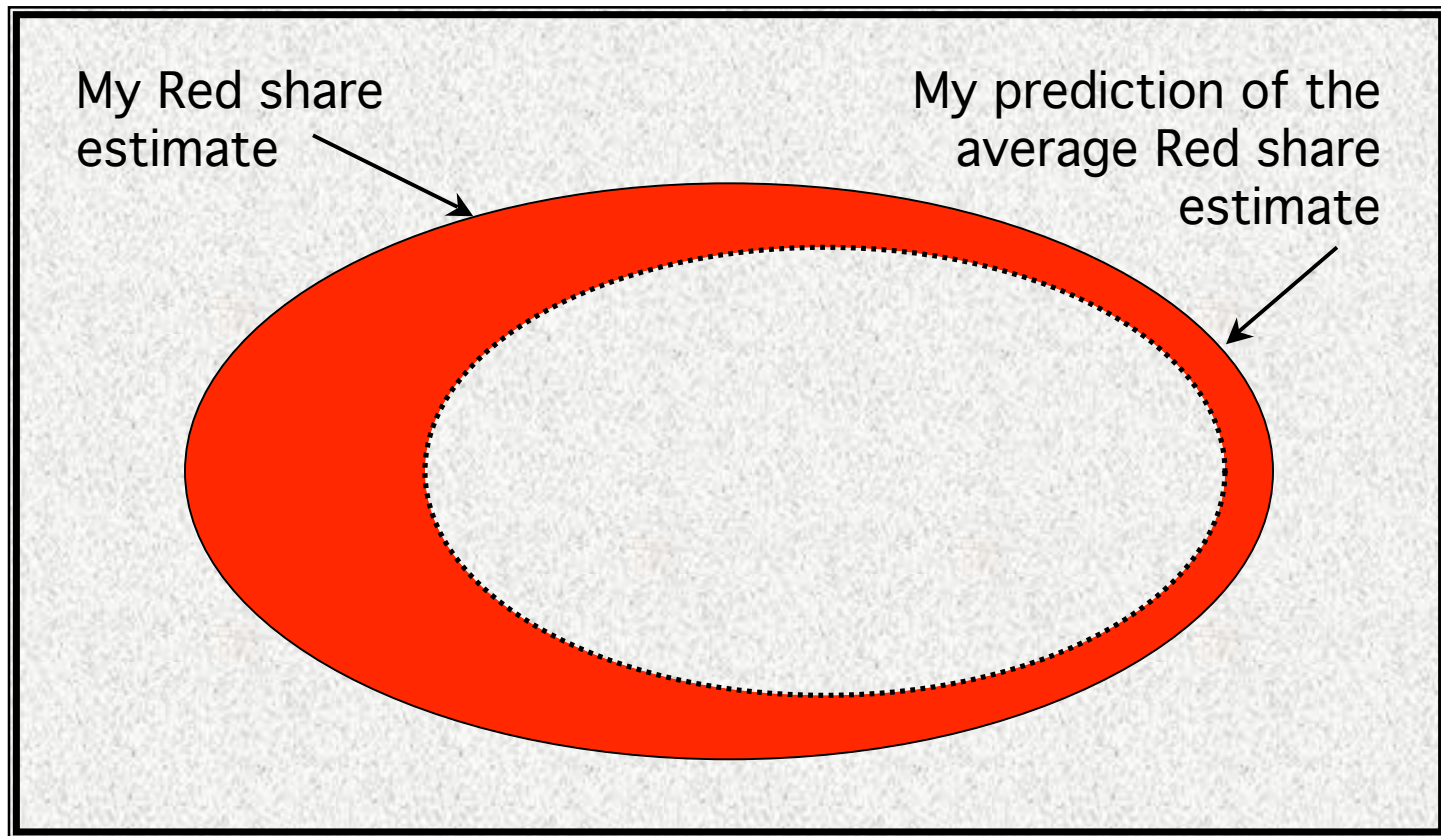
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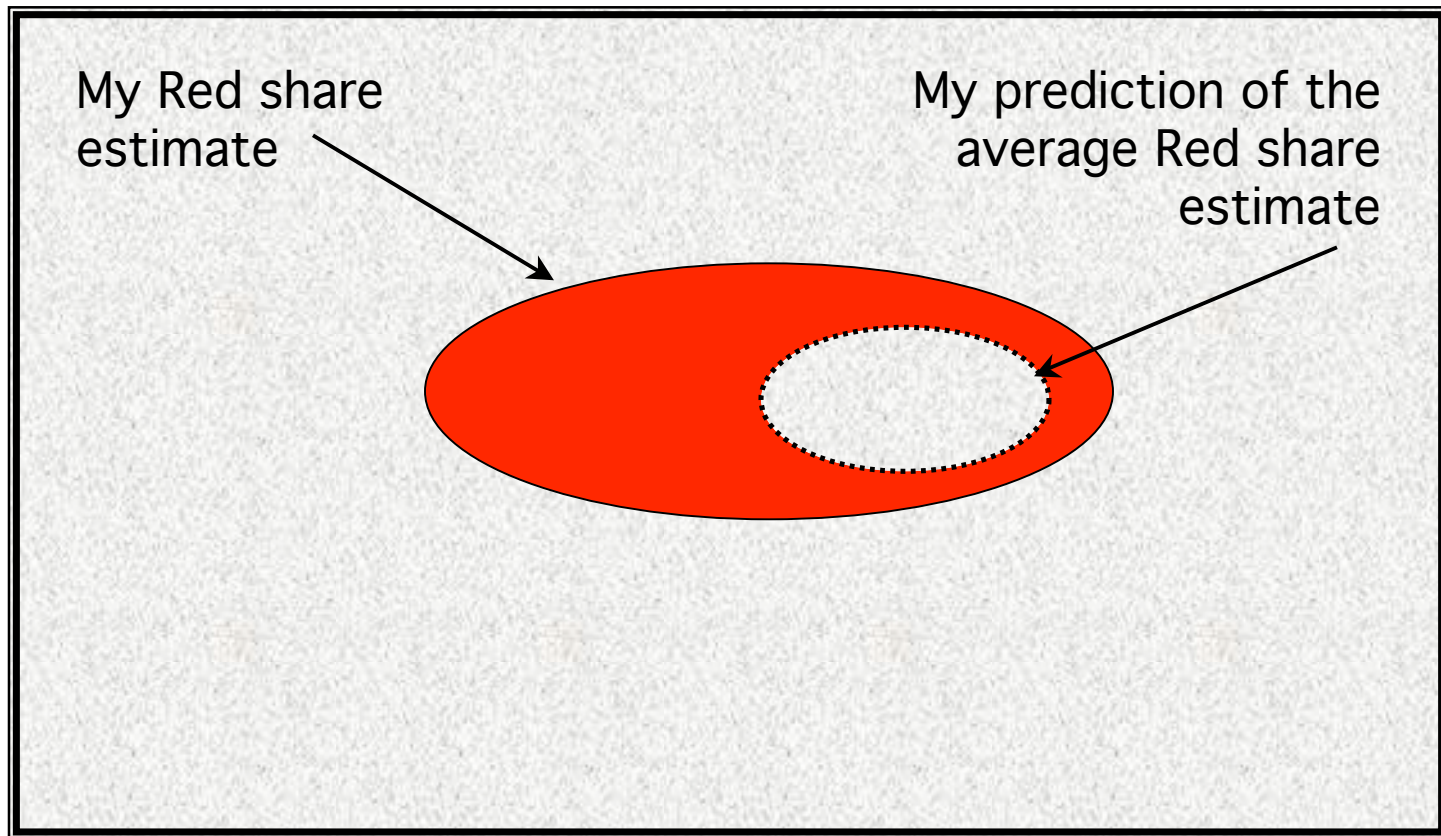
Hence, if I like Red I should believe that the share for Red will be underestimated



or, that Red will be ‘surprisingly popular’



The argument holds even if I know that my preferences are unusual



Proof strategy: Find an expression for expected score that lets you apply Jensen's inequality

$$\text{If } \phi(\omega) \neq \xi(\omega), \text{ then } \int_{\Omega} \phi(\omega) \log \phi(\omega) d\omega > \int_{\Omega} \phi(\omega) \log \xi(\omega) d\omega$$

Part I: Calculate (ex-post) information-score,
assuming true distribution is ω

$$\log \frac{\bar{x}_j}{\bar{y}_j} =$$

Assuming **actual** distribution is ω ,
the information score for j will be:

$$\log \frac{\bar{x}_j}{\bar{y}_j} = \sum_{k=1}^m p(t^s=k | \omega) \log \frac{p(t^r=j | \omega)}{p(t^r=j | t^s=k)}$$

$$\log \bar{x}_j = \log p(t^r=j | \omega)$$

$$\log \bar{y}_j = \sum_{k=1}^m p(t^s=k | \omega) \log E\{\bar{x}_j | t^s = k\}$$

$$= \sum_{k=1}^m p(t^s=k | \omega) \log p(t^r=j | t^s=k)$$

just a factor of 1

$$\begin{aligned} \log \frac{\bar{x}_j}{\bar{y}_j} &= \sum_{k=1}^m p(t^s=k|\omega) \log \frac{p(t^r=j|\omega)}{p(t^r=j|t^s=k)} \\ &= \sum_{k=1}^m p(t^s=k|\omega) \log \frac{p(t^r=j|\omega)}{p(t^r=j|t^s=k)} \frac{p(t^s=k|\omega)}{p(t^s=k|\omega)} \end{aligned}$$

Conditional independence

$$\begin{aligned}\log \frac{\bar{x}_j}{\bar{y}_j} &= \sum_{k=1}^m p(t^s=k | \omega) \log \frac{p(t^r=j | \omega)}{p(t^r=j | t^s=k)} \\&= \sum_{k=1}^m p(t^s=k | \omega) \log \frac{p(t^r=j | \omega) p(t^s=k | \omega)}{p(t^r=j | t^s=k) p(t^s=k | \omega)} \\&= \sum_{k=1}^m p(t^s=k | \omega) \log \frac{p(t^r=j, t^s=k | \omega)}{p(t^r=j | t^s=k) p(t^s=k | \omega)} \quad (\text{Conditional Ind.})\end{aligned}$$

Information score for j measures how much another person's beliefs about actual ω are changed by learning that someone else has opinion j

$$\begin{aligned}
 \log \frac{\bar{x}_j}{\bar{y}_j} &= \sum_{k=1}^m p(t^s=k | \omega) \log \frac{p(t^r=j | \omega)}{p(t^r=j | t^s=k)} \\
 &= \sum_{k=1}^m p(t^s=k | \omega) \log \frac{p(t^r=j | \omega) p(t^s=k | \omega)}{p(t^r=j | t^s=k) p(t^s=k | \omega)} \\
 &= \sum_{k=1}^m p(t^s=k | \omega) \log \frac{p(t^r=j, t^s=k | \omega)}{p(t^r=j | t^s=k) p(t^s=k | \omega)} && \text{(Conditional Ind.)} \\
 &= \sum_{k=1}^m p(t^s=k | \omega) \log \frac{p(\omega | t^r=j, t^s=k)}{p(\omega | t^s=k)}. && \text{(Bayes' Rule)}
 \end{aligned}$$

Part II: Calculate ex-ante expected information-score, conditional on giving answer j to opinion i

$$E\left\{ \log \frac{\bar{x}_j}{\bar{y}_j} \mid t^r = i \right\} = \int_{\Omega} \sum_{k=1}^m p(\omega \mid t^r = i) p(t^s = k \mid \omega) \log \frac{p(\omega \mid t^r = j, t^s = k)}{p(\omega \mid t^s = k)} d\omega$$

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 &= \int_{\Omega} \sum_{k=1}^m \frac{p(t^r = i \mid \omega) p(\omega) p(t^s = k \mid \omega)}{p(t^r = i)} \log \frac{p(\omega \mid t^r = j, t^s = k)}{p(\omega \mid t^s = k)} d\omega \quad (\text{Bayes' Rule})
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 &= \int_{\Omega} \sum_{k=1}^m \frac{p(t^r = i, t^s = k \mid \omega) p(\omega)}{p(t^r = i)} \log \frac{p(\omega \mid t^r = j, t^s = k)}{p(\omega \mid t^s = k)} d\omega \quad (\text{Conditional Ind.})
 \end{aligned}$$

This is the desired form:
 maximized iff: $\phi = \xi$, i.e., $j=i$

$$\int_{\Omega} \phi(\omega) \log \xi(\omega) d\omega$$

$$\begin{aligned}
 E\left\{ \log \frac{\bar{x}_j}{y_j} \mid t^r = i \right\} &= \int_{\Omega} \sum_{k=1}^m p(\omega \mid t^r = i) p(t^s = k \mid \omega) \log \frac{p(\omega \mid t^r = j, t^s = k)}{p(\omega \mid t^s = k)} d\omega \\
 &= \int_{\Omega} \sum_{k=1}^m \frac{p(t^r = i \mid \omega) p(\omega) p(t^s = k \mid \omega)}{p(t^r = i)} \log \frac{p(\omega \mid t^r = j, t^s = k)}{p(\omega \mid t^s = k)} d\omega \quad (\text{Bayes' Rule}) \\
 &= \int_{\Omega} \sum_{k=1}^m \frac{p(t^r = i, t^s = k \mid \omega) p(\omega)}{p(t^r = i)} \log \frac{p(\omega \mid t^r = j, t^s = k)}{p(\omega \mid t^s = k)} d\omega \quad (\text{Conditional Ind.}) \\
 &= \sum_{k=1}^m p(t^s = k \mid t^r = i) \int_{\Omega} p(\omega \mid t^r = i, t^s = k) \log \frac{p(\omega \mid t^r = j, t^s = k)}{p(\omega \mid t^s = k)} d\omega \quad (\text{Bayes' Rule})
 \end{aligned}$$


This is the desired form:
 maximized iff: $\phi = \xi$, i.e., $j = i$

$$\int_{\Omega} \phi(\omega) \log \xi(\omega) d\omega$$

doesn't depend on j

$$= \sum_{k=1}^m \cancel{p(t^s=k | t^r=i)} \int_{\Omega} p(\omega | t^r=i, t^s=k) \log \frac{p(\omega | t^r=j, t^s=k)}{\cancel{p(\omega | t^s=k)}} d\omega \quad (\text{Bayes' Rule})$$


Theorem 1 (Prelec, 2004)

Truthtelling is Bayes Nash Eq in a large sample

Collective truthtelling means that all answers and predictions are truthful, and consistent with Bayes' rule.

- Theorem 1A Truthtelling is a strict Bayesian Nash equilibrium in a countably infinite sample.
- Theorem 1C A respondent's BTS score in the truthtelling equilibrium equals the log posterior probability she assigns to the actual distribution of signals, ω , plus a budget balancing constant:

$$u^r = \log p(\omega | t^r) + b(\omega)$$

Hence, the difference between respondents' scores is a log-likelihood ratio,

$$u^r - u^s = \log p(\omega | t^r) - \log p(\omega | t^s).$$

Comparing BTS and prediction markets

- Common characteristics:
 - incentive compatible (truthtelling is optimal)
 - zero-sum (budget balance)
 - non-democratic aggregation of information, favoring informed participants (experts)
- Differences
 - BTS is one-shot, markets are dynamic
 - BTS is not restricted to verifiable events (claims)

BTS (Bayesian Truth Serum) Scoring

Your answers in this survey will be scored with a new mathematical algorithm called the Bayesian Truth Serum.

This algorithm was published in the journal Science on October 14, 2004.



The algorithm scores both your personal answers to each question, and your percentage predictions about the answers of others.

The maximum score is 800, just like on an SAT or GRE test. The lowest score is 400. The higher your score, the more chance you have of getting the bonus.

REPORTS

A Bayesian Truth Serum for Subjective Data

Douglas Riecke

Intuitive judgments, or so-called information content as for evidence and priors, are problems because there are no public criteria for assessing judgmental truthfulness. I present a scoring method for eliciting truth-biased subjective data in situations where objective truth is unknown. The method assigns high scores to the most common answers to the question that are more common than collectively paid out, with predictions drawn from the same population. The simplest adjustment for paying out errors comes at the loss of common, truthful answers, because maximum responsiveness can be in opposition to the false that their answer supports a reality view.

Subjective judgment from expert and lay witnesses is common in all human knowledge. Sources of behavior, attitude, and intention are in virtually every field of political science, psychology, sociology, and economics [1]. Subjective expert judgment drives environmental risk analysis, business forecasts, historical diagnosis, and artistic and legal interpretations [2].

The value of subjective data is limited by its quality in the source—a thought process of an individual produces output. Quality is most generally the relevance of the data [3], as if their answers were being elicited by an evaluation source who knows the truth [3]. This is the relation with tests of objective knowledge, where accuracy is defined as agreement with a known answer key, or in the case of humans, an desirable outcome [4]. Such evaluations are easily appropriate to social sciences, because one attempts to estimate to predict a particular definition of truth, even if one were possible [5].

But, if present a method of eliciting subjective information, designed for situations where objective truth is unknown or practically unobtainable [6]. The method consists of an "information content" test that induces truthful answers from a sample of interest [3]. (Answers) express maximum-minimum responsiveness. Unlike other Bayesian elicitation techniques [7-9], the method does not assume that the researcher knows the individual's information state that different responses. Hence, it can be applied to previously elicited questions, to a researcher who is a complete outsider to the domain. Unlike earlier approaches to "test your beliefs" in accuracy [10, 11], it is

Reprints ordered only, it does not privilege the common answer. Hence there is no reason for respondents to bias their answers toward the likely group mean. Truthful responding remains the strong reason for why someone who does not know cannot optimaize for success.

Instead of more common, a truth criterion, we method assigns high scores to answers that are more common than collectively paid out, with predictions from the same population that generates the answers. This method is "empirically neutral," and the associated numerical index, is called an information score. This adjustment in the target criterion increases the bias in favor of, or counteracts methods that break the giving-false-between (paid and unpaid) options.

The scoring is above the level of a simple question. For example, one might ask, "Is there a cure for prostate cancer that increases life expectancy past the 2000-2004 population probability score?" Will you vote in the next presidential election (Daily/Probably/Probably Not/Definitely Not) (see how you had more than 20 normal patients vote on this question [12]).

Each respondent provides a personal belief and an information content response distribution of answers (i.e., the fraction of people making each answer). Predictions are scored for accuracy, that is, for how well they match the overall population. The personal answers, which are the main output of interest, are scored for being responsive answers. An answer collected by 10% of the population against a predicted frequency of 5% means the answer is responsive and would receive a 1.6 information score, if predictions averaged 20%. It would be a negatively responsive answer, and hence not included in a true score.

The negatively responsive criterion provides an essential insight into the Bayesian reasoning about population frequency. Namely, in most situations, one should expect that others will underestimate the true frequency of one's own opinion or personal characteristics. This implication is useful in the policy space because it suggests that higher probabilities of the occurrence of a given opinion or characteristic in the population should come from individuals who hold the opinion or characteristic, because holding the opinion constitutes a valid and favorable signal about its general probability [13, 14]. People who, for example, 100 times in their lifetime should be asked to do (30%-plus) likely estimates of the percentage of the population who shares this opinion, receive their own beliefs as an "information" sample of one's [15]. It follows that the Bayesian knows who have more to believe than that their own estimate of the population is high compared with others' estimates. And one can take that their population of people is underestimated by the population. Hence, one's own opinion is also the opinion that has the best chance of being negatively accurate.

The validity of the personally held answers depend on whether the personally held answers are more common than the population. For example, a male who has had more than 20 normal patients (prostate question) [16] may find that few fall into the prostate cancer category. Therefore, according to Bayesian reasoning, this doctor's judgment has personal accuracy higher percentage (i.e., the will be somewhat higher than the average of estimates collected from the population in a future study [17]). One can take that he had more than 20 normal patients to estimate that the general population, which includes prostate with his patients, will underestimate the prevalence of the prostate.

Such findings, individually collected information, provide a useful answer to the question of information content scores, assuming they represent a responding population. Hence, it is a Bayesian Truth Sermon [18]. It is also collectively oriented in the sense that it does not provide a positive a priori answer. The actual application of the method, one would not build respondents the confidence of scoring or explain the source of reliability. Hence, it would take the role of a truth test. One can assume that the method is not reported scores, and that in testing at their personal true answer they can find to ignore what other respondents might say. The optimization analysis confirms that under con-

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To maximize your BTS score you should try to make sure that:

- **your answers are authentic** — that they accurately reflect your true opinions and preferences
- **your predictions are your best % estimates of how others will answer.**

What is the actual scoring formula?

The formula is related to the mathematical theory of information. Let $\bar{x}_i$ be the actual % that choose answer i out of m answers, and $\bar{y}_i$ the geometric average prediction of that %. Then, a person who chooses answer k and predicts percentages $y_1, \dots, y_m$ gets a score of (A, B are optional constants):

$$A \left(\log \frac{\bar{x}_k}{\bar{y}_k} + \sum_{i=1}^m \bar{x}_i \log \frac{y_i}{\bar{x}_i} \right) + B$$

Do I need to understand the formula or how it works?

No. Understanding the formula does not provide any advantage. It is a mathematical theorem that no matter what your opinion, giving an honest answer will maximize your expected score. That is all you need to know.

Is BTS a lie detector?

No. BTS cannot tell if any single answer is honest. However, honest answers maximize your chance of getting a high BTS score.

This is exactly the same situation as with an SAT or GRE test.

With the SAT, your best strategy is to give answers that you believe are correct. This way you achieve your personal maximum. If you answer carelessly or choose incorrect answers on purpose, your score will go down. But, a low score does not prove that a person is dishonest. They simply might not know the correct answer.

With a BTS test, also, your best strategy is to give answers that accurately reflect your opinions. This way you will achieve your personal maximum. But, a low score does not prove that a person is dishonest.

What is the main difference between BTS and SAT-type scoring?

SAT scores reflect your aptitude and the effort you put in the test.

BTS scores reflect three things: (1) your self-knowledge, (2) your effort, and (3) the statistical information value of your answers. BTS does not have a predetermined "correct answer" key. No one — including us! — knows in advance which answers will get higher scores.

Does it matter whether my opinion is unusual?

No. The scoring is neutral, not biased for or against unusual answers. You do not need to worry whether your answer is typical.

The logarithmic proper scoring rule rewards truthful probability estimates

Expert's true subjective probability of disaster = p

Expert announced probability of disaster = y

After the outcome is known, the expert receives a score:

$$\text{Score} = \begin{cases} K + \log y, & \text{if disaster} \\ K + \log (1-y), & \text{if no disaster} \end{cases}$$

Elementary theorem:

Truth-telling ($y=p$) maximizes expected score, which is:

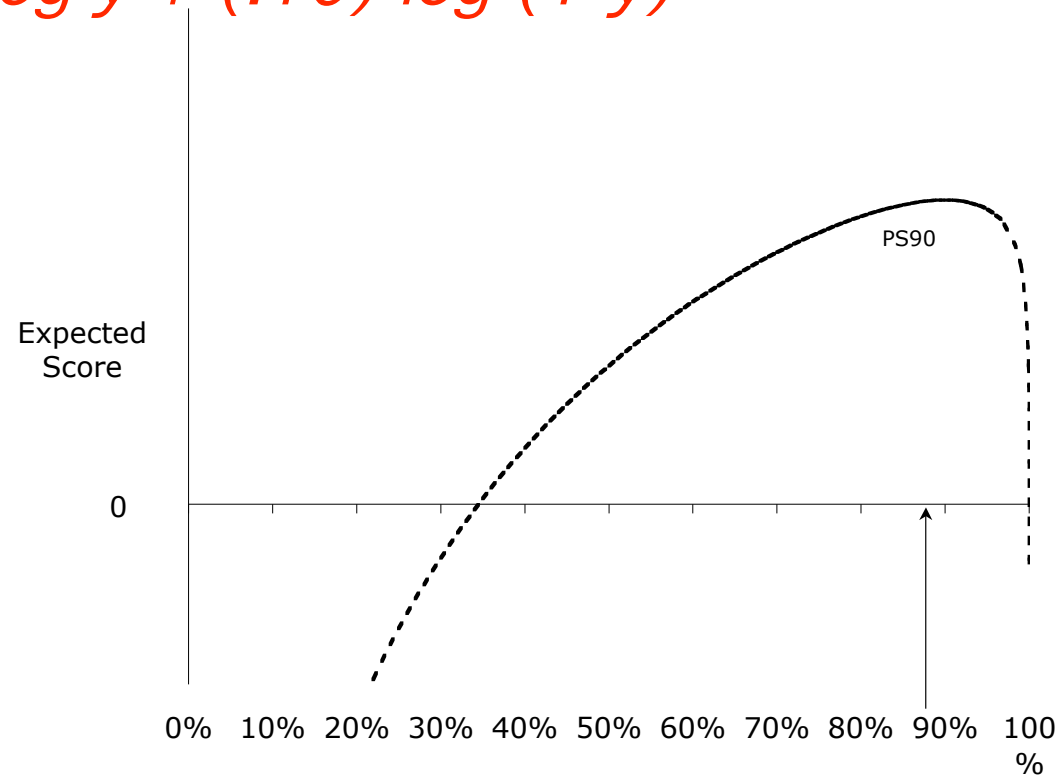
$$K + p \log y + (1-p) \log (1-y)$$

Imagine that expert has true $p = 90\%$
and calculates expected value for all y :

$$K + (.90)\log y + (.10) \log (1-y)$$

Imagine that expert has true $p = 90\%$
and calculates expected value for all y :

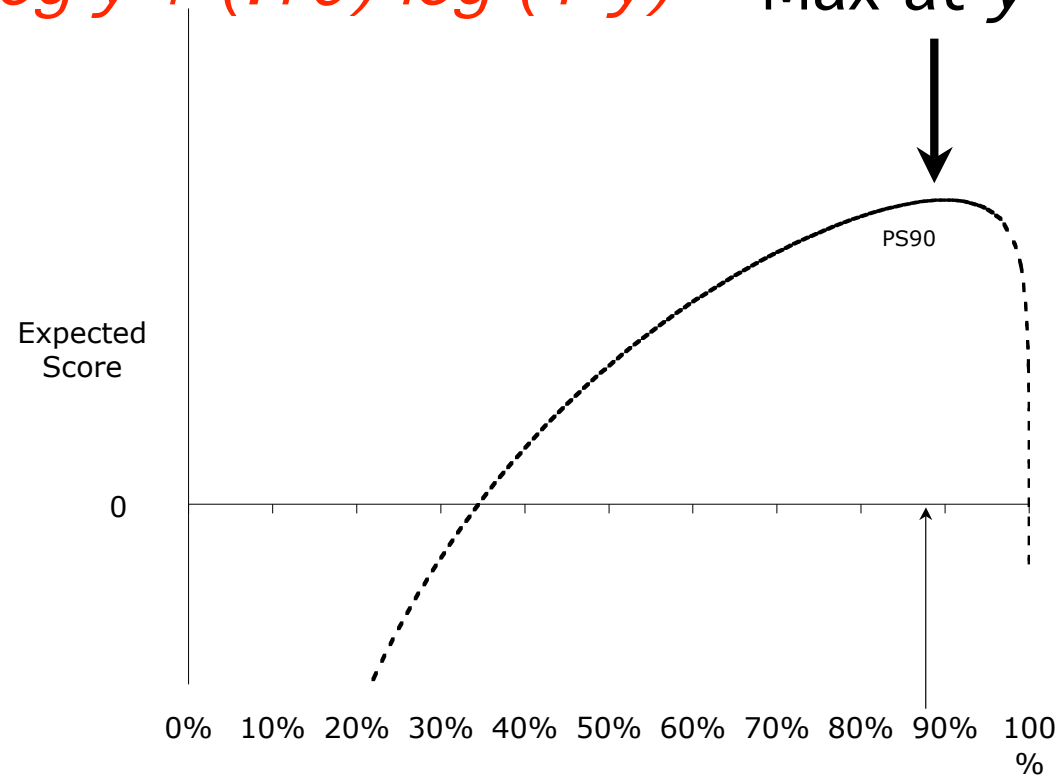
$$K + (.90)\log y + (.10)\log(1-y)$$



y = Reported probability of catastrophe by 2100

Imagine that an expert has true $p = 90\%$
and calculates expected value for all y :

$$K + (.90)\log y + (.10)\log(1-y) \quad \text{Max at } y = p = .90$$



y = Reported probability of catastrophe by 2100

For each product:

1. Indicate with an "X" how likely it is that you would buy the product sometime in the near future.
2. Estimate the % of women in this class who will mark each of the four answers to question 1 (the total across all 4 answers should be 100%).
3. Estimate the % of men in this class who will mark each of the four answers to question 1 (the total across all 4 answers should be 100%).

A. Portable Mini Cycle **retail price \$99.95**



Portable Mini Cycle tightens and tones legs and arms with adjustable resistance.

- Place this portable stationary bike on the floor and cycle to strengthen legs as you add shape and definition.
- Or place it on a tabletop and operate with your hands for firming up hard-to-tone muscles under upper arms.
- Turn the dial to adjust the resistance from a light workout to a rigorous one.
- Built-in computer with LCD shows speed, workout distance, workout time, total distance and estimated calories burned.

	Definitely Buy	Probably Buy	Probably Not Buy	Definitely Not Buy
You	_____	<u>X</u>	_____	_____
Women	<u>5</u> %	<u>15</u> %	<u>45</u> %	<u>35</u> %
Men	<u>0</u> %	<u>2</u> %	<u>18</u> %	<u>80</u> %

B. Motorized DVD Tower **retail price \$169.95**



Store 80 DVD cases in a space-saving motorized organizer that rotates 360° for quick, easy selection.

- Easy viewing at a comfortable, back-saving height.
- Ultra-bright LED lamp illuminates cases in a darkened room.
- Entire collection rotates 360° clockwise or counterclockwise.
- Occupies barely a square foot of floor space.

	Definitely Buy	Probably Buy	Probably Not Buy	Definitely Not Buy
You	_____	_____	_____	<u>X</u>
Women	<u>0</u> %	<u>0</u> %	<u>25</u> %	<u>75</u> %
Men	<u>10</u> %	<u>20</u> %	<u>30</u> %	<u>40</u> %

C. Rhythm Stix **retail price \$14.95**



Always wanted a drum set? Get a pair of Rhythm Stix and you've got a kit of percussive sounds!

- Switch on each drumstick and tap them on any hard surface to hear the realistic sounds of a professional-style drum kit.
- Built-in speakers blast out techno-tom-tom beats, crashing cymbals and spectacular snare sounds.
- A brilliant blue LED illuminates each time the tip of a stick strikes.
- Press "Rhythm" to enjoy hip-hop music along with your ultra-cool drumming.

	Definitely Buy	Probably Buy	Probably Not Buy	Definitely Not Buy
You	_____	_____	<u>X</u>	_____
Women	<u>5</u> %	<u>15</u> %	<u>40</u> %	<u>20</u> %
Men	<u>15</u> %	<u>20</u> %	<u>50</u> %	<u>15</u> %

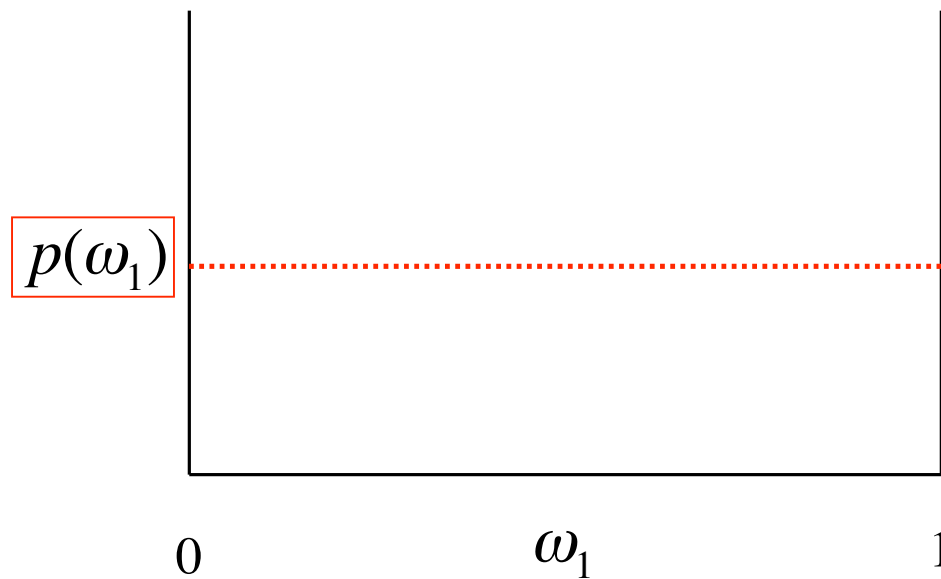
What is your gender?



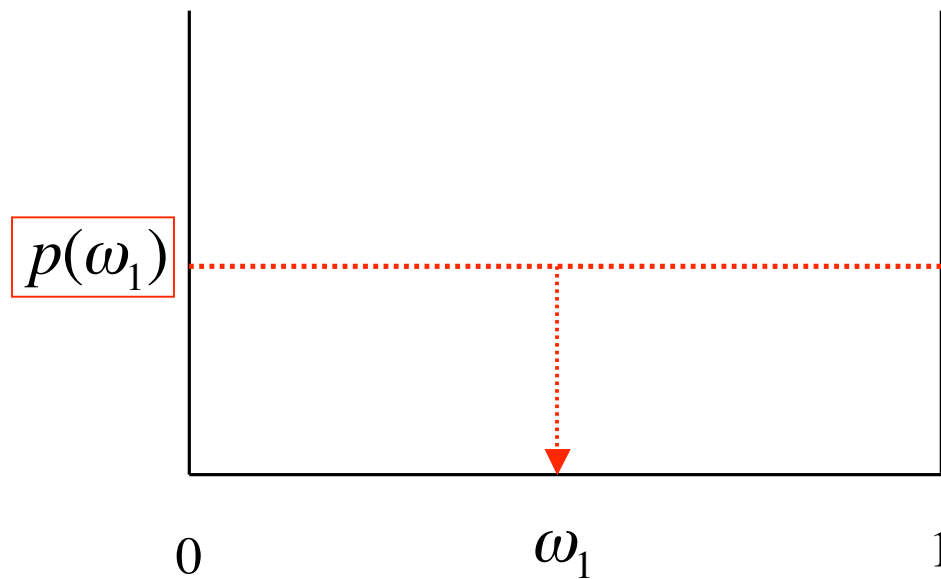
M

Prelec and Weaver, 2006

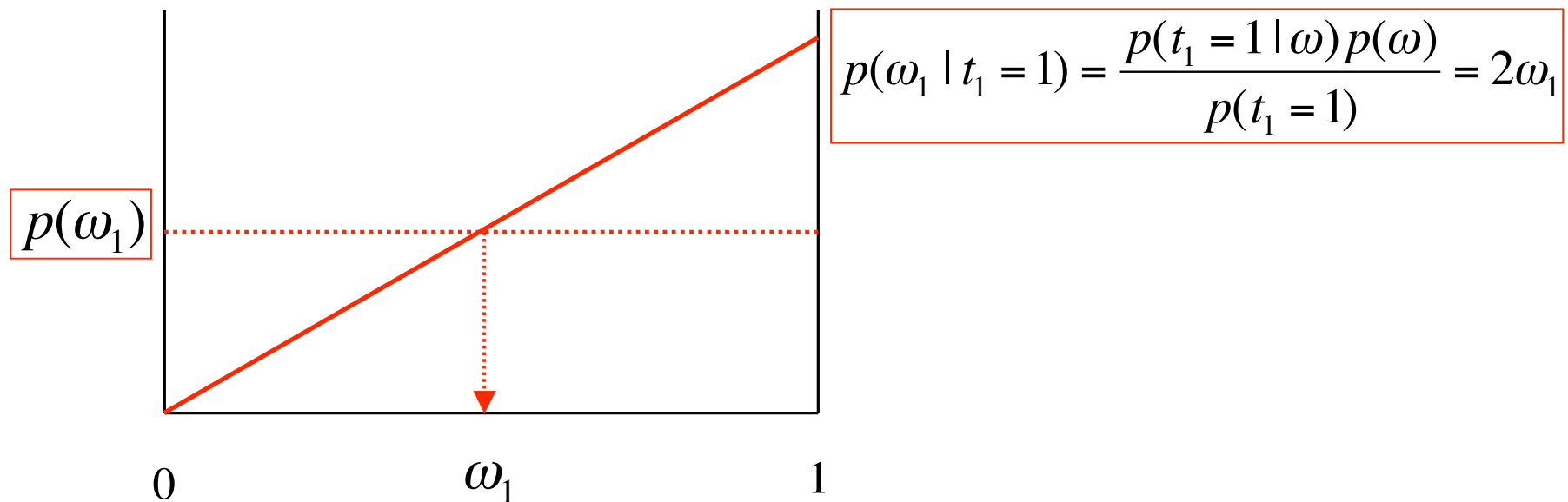
Example: A bag contains Red and White balls in unknown proportions, 1=Red, 2=White, $\omega = (\omega_1, \omega_2)$



Uniform prior (all proportions equally likely)
Prior expected frequency of Red = 0.5



==> Triangular posterior distribution of Red
conditional on drawing one Red ball



Posterior expected frequency = 0.67

